GEOMETRY QUEST · WORLD 1

# Points & Lines

**ANSWER SHEET · for the grown-up**

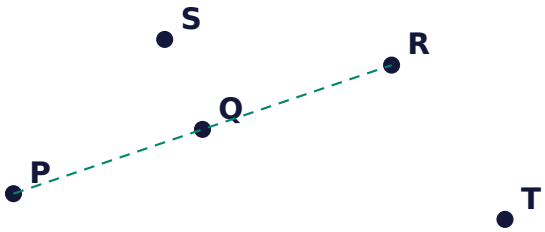
Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

**1** Name each one: line segment, ray or line.



**segment** **ray** **line**

**2** Three of these points are collinear (on the same straight line). Which three? Use a ruler to check.




**P, Q and R**

**3** Measure line segment AB with a ruler. How long is it?




**7.5** cm

**4** How many different straight lines can you draw through two points?

**exactly 1**

**5** Draw a line segment exactly 6 cm long. Label its ends C and D.

**Check: 6 cm, both ends labelled.**

**6** Real-life hunt: name something at home that is like a ray, and something like a line segment.

**e.g. ray: torch beam, sunbeam · segment: pencil, table edge**


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GEOMETRY QUEST · WORLD 2

# Angles

**ANSWER SHEET · for the grown-up**

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

**1** Write acute, right, obtuse, straight or reflex under each angle.



**acute** **right** **obtuse** **straight** **reflex**

**2** What angle do the hands of a clock make at each time? (Think: 360° ÷ 12 = 30° per hour.)

**3:00 →** **90** ° **6:00 →** **180** ° **2:00 →** **60** ° **4:00 →** **120** °

**3** Fill in the gaps.

Two angles that add up to 90° are called **complementary**

Two angles that add up to 180° are called **supplementary**

The complement of 35° is **55°**

The supplement of 110° is **70°**

**4** Turning. You face North. Which way are you facing after each turn (from North each time)?

a quarter turn clockwise (90°) **East**

a half turn (180°) **South**

270° clockwise **West**

**5** Skater challenge: a “540” is how many full turns?

**1½ turns**



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GEOMETRY QUEST · WORLD 3

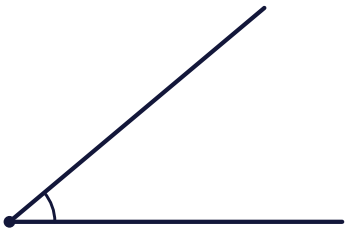
# Protractor Skills

**ANSWER SHEET · for the grown-up**

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

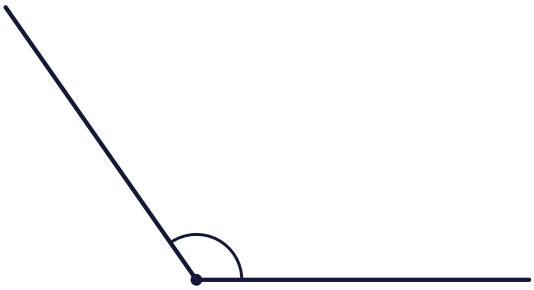
**1** Measure each angle with a protractor. Put the centre on the corner and count up from the 0 on one arm. (Drawn to size: print at 100%.)

**a)**



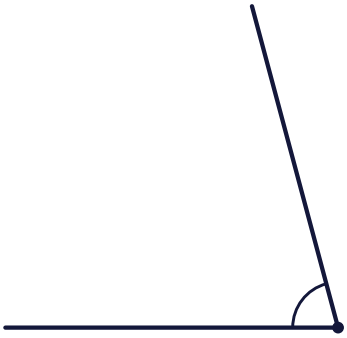
**40** °

**b)**



**125** °

**c)**



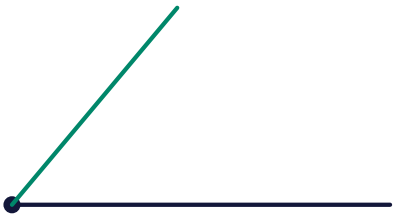
**75** °

**d)**



**160** °

**2** Use your protractor to draw these angles on the lines below. Mark the angle with an arc.



**a) 50°**

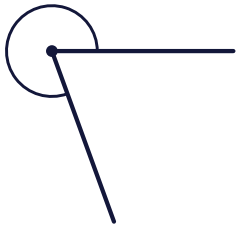


**b) 135°**



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**3** This is a reflex angle. Measure it. (Tip: measure the small angle, then take it away from 360°.)



**290** °

**(small angle 70°, 360 − 70 = 290)**

**4** Sam measured an angle and wrote 140°. But the angle is clearly smaller than a right angle. What went wrong, and what is the real size?

**Read the wrong scale. The angle is 180 − 140 = 40°.**


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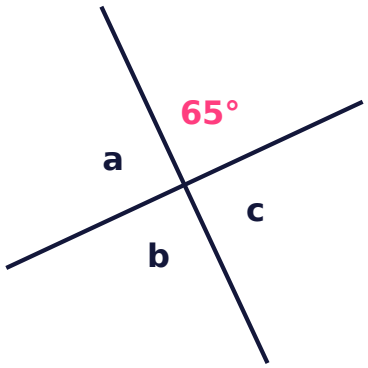
GEOMETRY QUEST · WORLD 4

# Angle Rules

**ANSWER SHEET · for the grown-up**

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

**1** Two straight lines cross. Find a, b and c. Give a reason for each. (Not drawn to scale.)

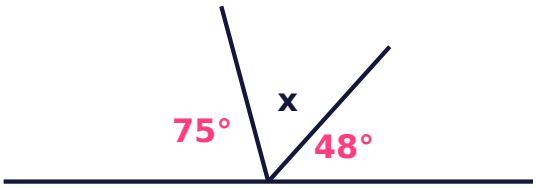



**a =** 115° (angles on a straight line)

**b =** 65° (vertically opposite)

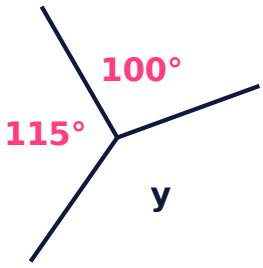
**c =** 115° (vertically opposite a)

**2** These angles sit on a straight line. Find x.




**x =** 57 °

**3** These angles meet at a point. Find y.




**y =** 145 °

**4** Four equal angles meet at a point. How big is each one?

90 °



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**5** Open a pair of scissors. Explain why the angle between the blades always equals the angle between the handles.

**Two straight lines cross, so vertically opposite angles are equal.**


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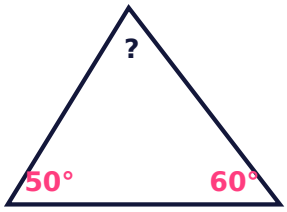
GEOMETRY QUEST · WORLD 5

# Triangles

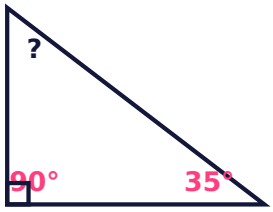
**ANSWER SHEET · for the grown-up**

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

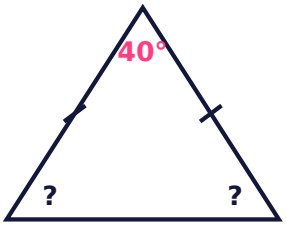
**1** Find the missing angle in each triangle. (Not drawn to scale.)




**70°**




**55°**




**70 each°**

**2** Classify each triangle by its sides (equilateral, isosceles, scalene) and its angles (acute, right, obtuse).

| | |
|---|---|
| sides 5, 5, 5 cm | **equilateral, acute** |
| angles 90°, 45°, 45° | **isosceles, right** |
| angles 120°, 40°, 20° | **scalene, obtuse** |

**3** Area of a triangle = ½ × base × height. Work these out.

| base 8 cm, height 5 cm | base 12 cm, height 7 cm | base 9 cm, height 4 cm |
|---|---|---|
| **20** cm² | **42** cm² | **18** cm² |

**4** An equilateral triangle has sides of 7 cm. What is its perimeter?

**21** cm

**5** Experiment: draw any big triangle. Measure its three angles with a protractor and add them up. What do you get?

**About 180° (within a few degrees because of measuring).**



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GEOMETRY QUEST · WORLD 6

# Polygons

**ANSWER SHEET · for the grown-up**



Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

**1** Name the polygon with this many sides.

| 5 sides | 6 sides | 8 sides | 12 sides |
|---|---|---|---|
| **pentagon** | **hexagon** | **octagon** | **dodecagon** |

**2** Angle total = (sides − 2) × 180°. Find the total of the inside angles.

| hexagon | octagon | decagon |
|---|---|---|
| **720** ° | **1080** ° | **1440** ° |

**3** Each inside angle of a regular polygon = angle total ÷ number of sides. Find one angle of a regular pentagon and a regular hexagon.

pentagon **108** ° **hexagon 120** °

**4** True or false?

| | |
|---|---|
| A square is a rectangle. | **true** |
| Every rectangle is a square. | **false** |
| A rhombus has four equal sides. | **true** |
| A trapezium has two pairs of parallel sides. | **false** |

**5** Find: (a) the area of a parallelogram with base 9 cm and height 4 cm; (b) the perimeter of a 7 cm by 3 cm rectangle.

(a) **36** cm² **(b) 20** cm

**6** Polygon hunt: find three polygons at home or outside. Name each one and say where you found it.

**Any sensible answers, e.g. hexagon: nut on a bolt.**



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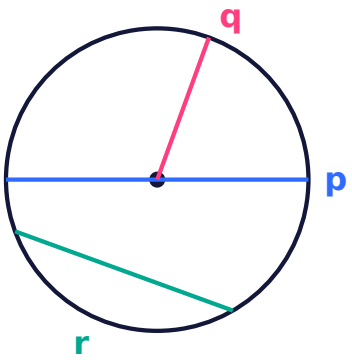
GEOMETRY QUEST · WORLD 7

# Circles & π

**ANSWER SHEET · for the grown-up**

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

**1** Which line is the radius, which is the diameter, and which is a chord?




radius: **q**

diameter: **p**

chord: **r**

**2** Fill in: a radius of 4 cm means a diameter of ___ cm. A diameter of 15 cm means a radius of ___ cm.

**8** cm **7.5** cm

**3** Circumference = π × diameter. Use π ≈ 3.14. Find the circumference when the diameter is 10 cm, and when it is 50 cm.

**31.4** cm **157** cm

**4** Area = π × radius × radius. Find the area when the radius is 5 cm, and when it is 10 cm.

**78.5** cm² **314** cm²

**5** A bike wheel is 70 cm across. How far does the bike travel in 10 full turns of the wheel? Give your answer in metres.

**≈ 22 m (2198 cm)**

**6** Kitchen experiment: wrap string round a tin, measure it, then measure across the tin. Divide the first by the second.

Around: _____

Across: _____

Around ÷ across = **≈ 3.1**


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GEOMETRY QUEST · WORLD 8

# 3D Shapes & Nets

**ANSWER SHEET · for the grown-up**

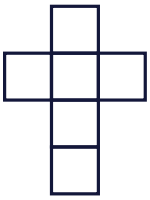
Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

**1** Complete the table. Then check that faces + vertices − edges = 2 for every shape.

| shape | faces | edges | vertices | F + V − E |
|---|---|---|---|---|
| cube | **6** | **12** | **8** | **2** |
| cuboid | **6** | **12** | **8** | **2** |
| triangular prism | **5** | **9** | **6** | **2** |
| square-based pyramid | **5** | **8** | **5** | **2** |
| tetrahedron | **4** | **6** | **4** | **2** |

**2** Which of these nets fold up into a cube? Tick them. (Imagine folding, or copy them onto squared paper and try.)

**A**



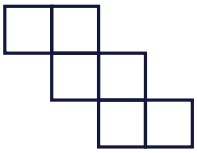
**yes**

**B**



**no**

**C**



**yes**

**D**



**no**

**3** Name the 3D shape.

I have 6 square faces. **cube**

1 square face + 4 triangles meeting at a point. **square-based pyramid**

I have 2 flat circles and 1 curved surface. **cylinder**

I have one curved surface and no edges at all. **sphere**

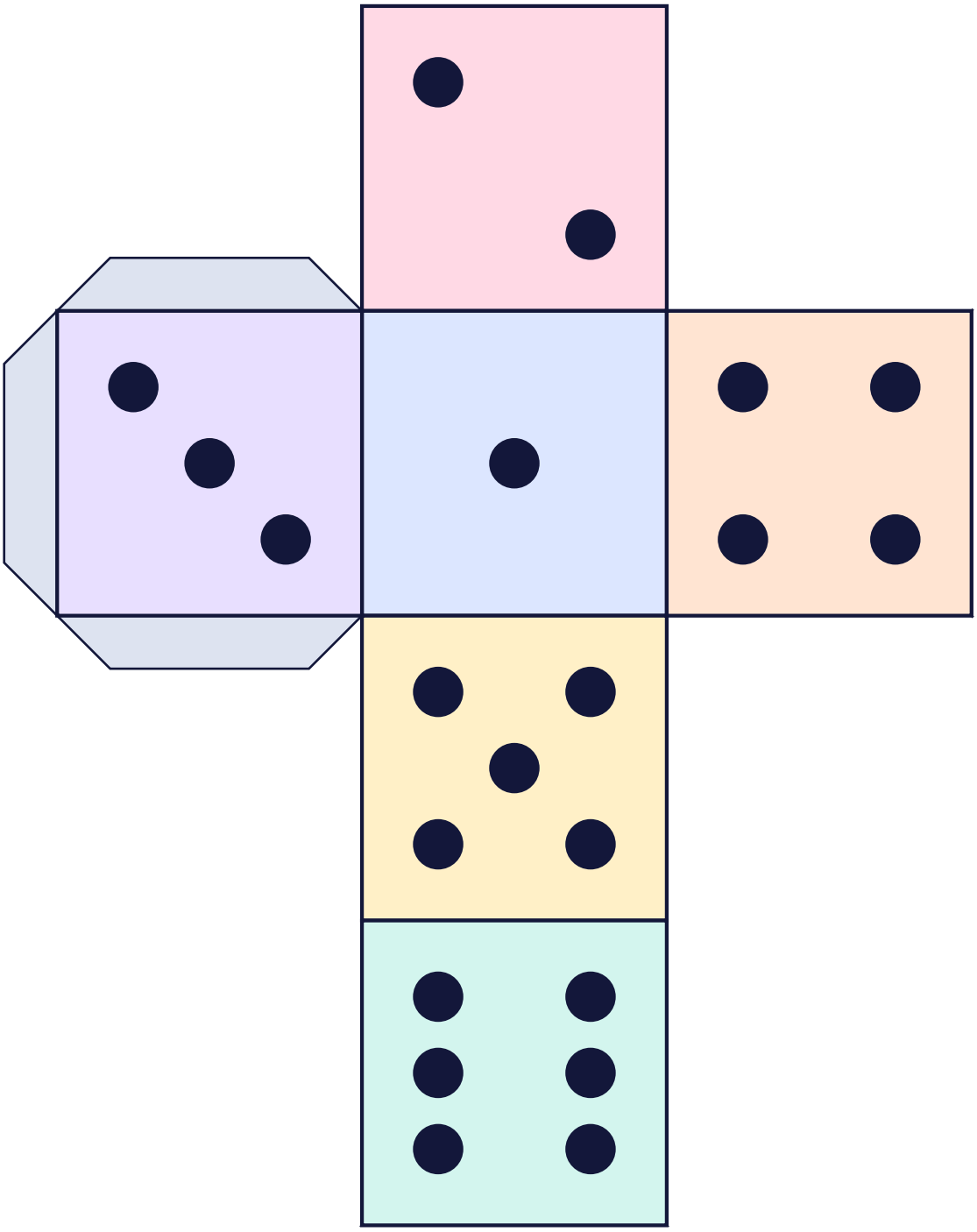
**4** Cut out the dice net on the next page, fold along the lines and glue the tabs. Opposite faces of a real dice add up to 7. Do yours?

**Yes: 1↔6, 2↔5, 3↔4 end up opposite.**



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## Cut out, fold along every line, glue the grey tabs. Build your own dice!




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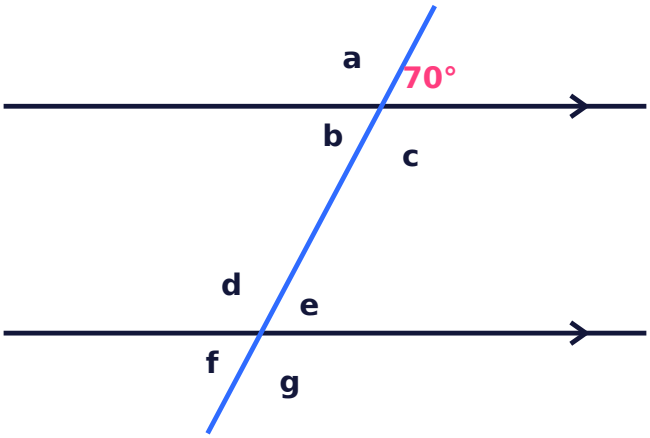
GEOMETRY QUEST · WORLD 9

# Parallel Lines

**ANSWER SHEET · for the grown-up**

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

**1** The two rails are parallel. One angle is 70°. Find angles a to g. (Not drawn to scale.)




**a =** 110°
**b =** 70°
**c =** 110°
**d =** 110°
**e =** 70°
**f =** 70°
**g =** 110°

**2** Use the diagram. Which kind of angle pair is each one: corresponding (F), alternate (Z) or co-interior (C)?

70° and e — **corresponding (F)**
b and e — **alternate (Z)**
b and d — **co-interior (C)**

**3** Two co-interior angles: one is 115°. What is the other?

**65** °

**4** Parallel hunt: find four pairs of parallel lines at home (think shelves, stairs, window frames).

**Any sensible answers.**


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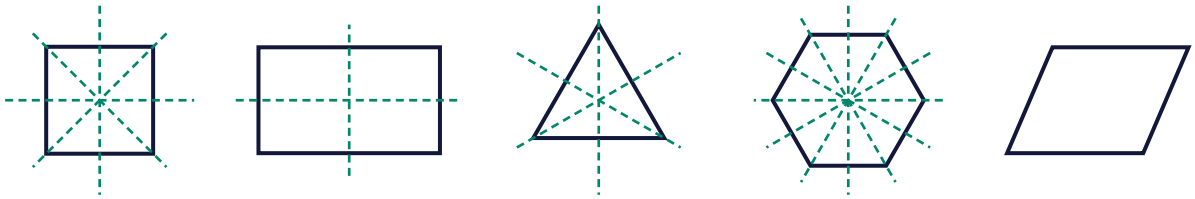
GEOMETRY QUEST · WORLD 10

# Symmetry

**ANSWER SHEET · for the grown-up**

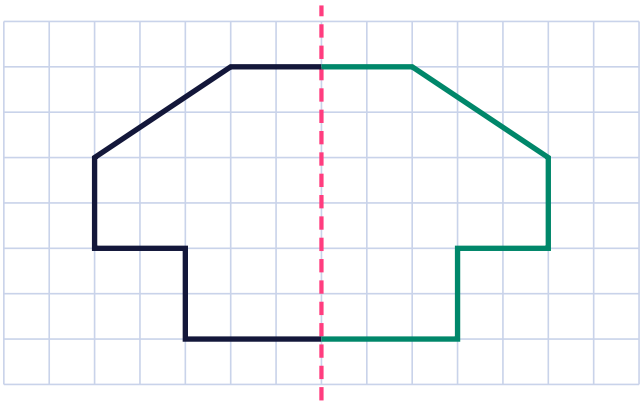
Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

**1** Draw all the lines of symmetry on each shape. How many does each have?



| square | rectangle | equilateral △ | reg. hexagon | parallelogram |
|---|---|---|---|---|
| **4** | **2** | **3** | **6** | **0** |

**2** Complete the picture so the dashed line is a line of symmetry.



**3** What is the order of rotational symmetry? (How many times does it fit onto itself in one full turn?)

| letter S | letter H | equilateral △ | square | recycling sign |
|---|---|---|---|---|
| **2** | **2** | **3** | **4** | **3** |

**4** Circle the capital letters that have a vertical line of symmetry: A B F H J M P T U W Y Z

**A H M T U W Y**



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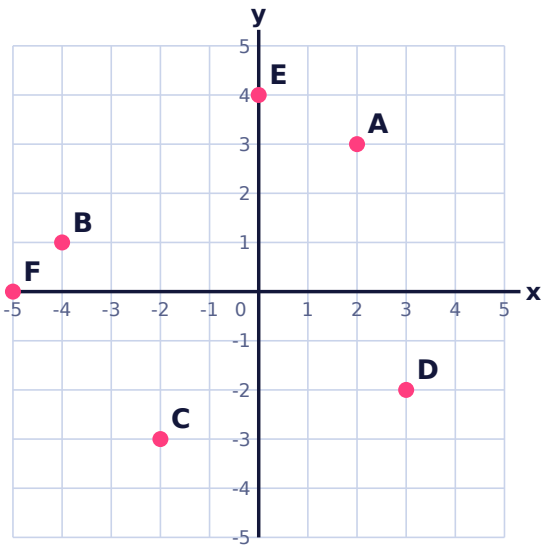
GEOMETRY QUEST · WORLD 11

# Coordinates

**ANSWER SHEET · for the grown-up**

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

**1** Write the coordinates of each point. Remember: across first, then up (x, y).




**A = ( 2 , 3 )**

**B = ( -4 , 1 )**

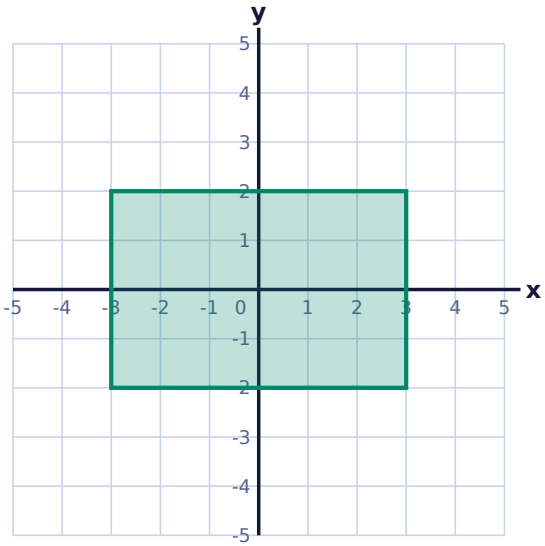
**C = ( -2 , -3 )**

**D = ( 3 , -2 )**

**E = ( 0 , 4 )**

**F = ( -5 , 0 )**

**2** Plot these points and join them in order: (−3, −2), (3, −2), (3, 2), (−3, 2), back to the start. What shape is it?




Shape: **rectangle**

**3** A square has corners at (1, 1), (4, 1) and (4, 4). Where is the fourth corner?

**(1, 4)**



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**4** Which quarter (quadrant) of the grid is each point in: top-left, top-right, bottom-left or bottom-right?

**(−2, 5)** top-left **(4, −1)** bottom-right **(−3, −3)** bottom-left


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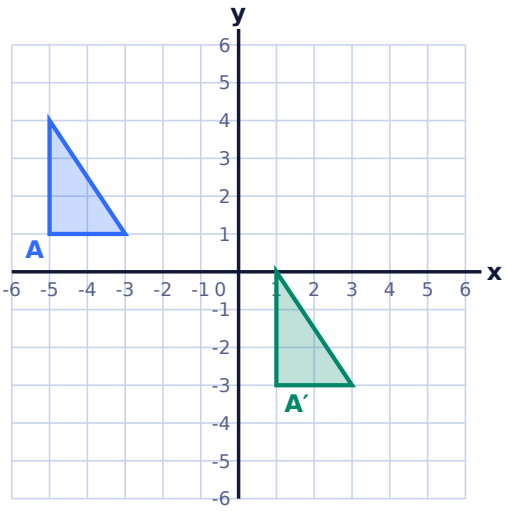
GEOMETRY QUEST · WORLD 12

# Transformations

**ANSWER SHEET · for the grown-up**

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

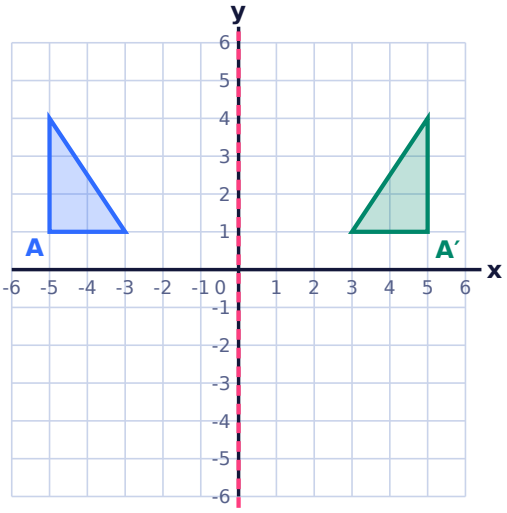
**1** Translate the triangle 6 squares right and 4 squares down. Label the image A′.




Write the coordinates of A′:

**(1, -3)**

**2** Reflect the triangle in the y-axis (the pink mirror line).



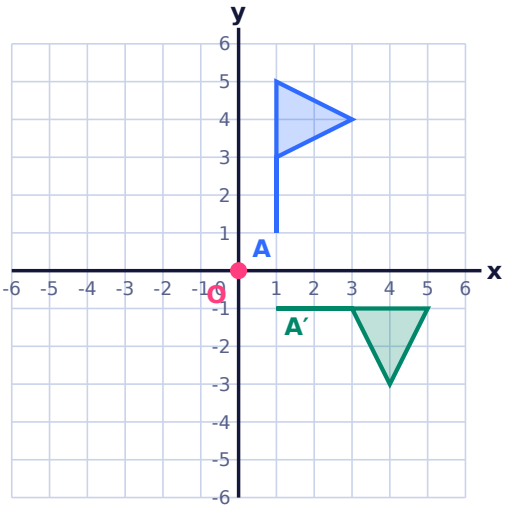

Write the coordinates of A′:

**(5, 1)**



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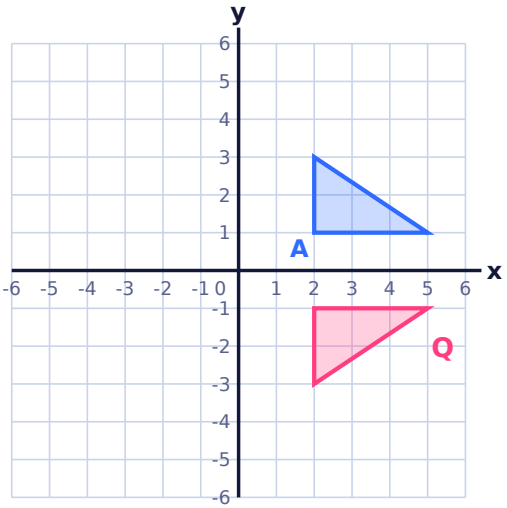
**3** Rotate the flag 90° clockwise about the origin O. (Tracing paper helps!)




Write the coordinates of A´:

**(1, -1)**

**4** Shape A was moved onto shape Q with one transformation. Describe it fully.




Describe the transformation:

**Reflection in the x-axis**

**5** After a translation, reflection or rotation, what stays the same about the shape?
Tick: size ☐ shape ☐ angles ☐ position ☐

**size, shape and angles (it is congruent). Position changes.**


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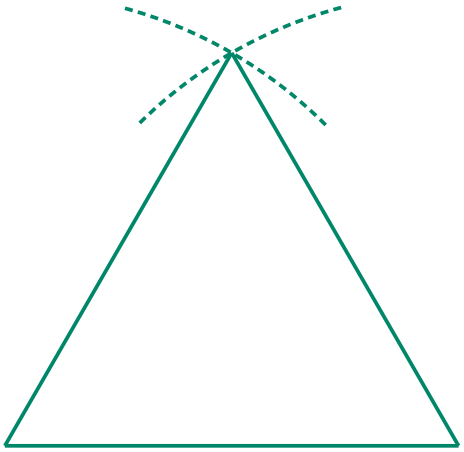
GEOMETRY QUEST · WORLD 13

# Constructions

**ANSWER SHEET · for the grown-up**

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

**1** Construct an equilateral triangle with sides of 6 cm. Only measure the first side; use your compass for the rest. Leave your arcs showing!



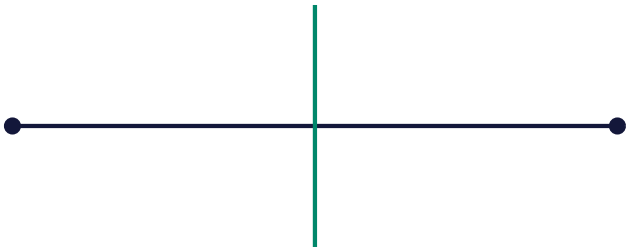
1. Draw AB = 6 cm.
2. Compass on A, open to B. Draw an arc.
3. Same opening, compass on B. Draw an arc.
4. Where the arcs cross is C. Join AC and BC.

**Check: all sides 6 cm, all angles 60°.**

**2** Can these three sticks make a triangle? Write yes, no, or flat.

| 3, 4, 5 cm | 2, 2, 5 cm | 5, 5, 10 cm | 6, 7, 8 cm |
|---|---|---|---|
| yes | no | flat | yes |

**3** Bisect (cut exactly in half) this 8 cm line with a compass: open it more than halfway, draw arcs above and below from each end, and join the crossings.



**Check: 4 cm each side**



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**4** Construct a triangle with sides 5 cm, 6 cm and 7 cm. (Draw the 7 cm side, then use arcs of 5 cm and 6 cm from its ends.)

**Check the three sides with a ruler: 5, 6 and 7 cm.**


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GEOMETRY QUEST · WORLD 14

# Congruence

**ANSWER SHEET · for the grown-up**



Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

**1** Two triangles share these facts. Which rule proves they are congruent: SSS, SAS, ASA, AAS, RHS or “not enough”?

| | |
|---|---|
| all three sides match | **SSS** |
| two sides and the angle between them match | **SAS** |
| two angles and the side between them match | **ASA** |
| a right angle, the hypotenuse and one other side match | **RHS** |
| all three angles match | **not enough** |
| two sides and an angle NOT between them match | **not enough** |

**2** △ABC ≅ △PQR. AB = 5 cm, BC = 7 cm and angle A = 40°. Find PQ, QR and angle P.

**PQ =** **5 cm** **QR =** **7 cm** **∠P =** **40°**

**3** Draw two triangles that both have angles of 50°, 60° and 70° but are different sizes. What does this show?

**Matching angles (AAA) is not enough: same shape, different size (similar).**

**4** Why does a garden gate have a diagonal piece of wood across it?

**The brace makes triangles, which are rigid (SSS), so the gate can’t sag.**



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GEOMETRY QUEST · WORLD 15

# Pythagoras & √2

**ANSWER SHEET · for the grown-up**

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

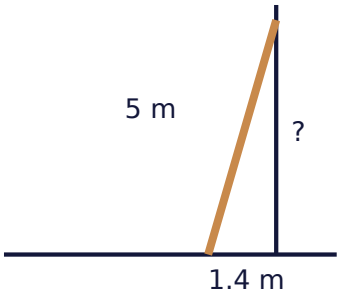
**1** Find the hypotenuse (longest side) of each right-angled triangle. $a^2 + b^2 = c^2$.

| 3 and 4 | 5 and 12 | 8 and 15 | 9 and 12 |
|---|---|---|---|
| **5** | **13** | **17** | **15** |

**2** Find the missing shorter side.

| hypotenuse 10, side 6 | hypotenuse 13, side 5 | hypotenuse 25, side 7 |
|---|---|---|
| **8** | **12** | **24** |

**3** A 5 m ladder leans on a wall. Its foot is 1.4 m from the wall. How high up the wall does it reach?




**4.8** m

**√(25 − 1.96) = √23.04 = 4.8**

**4** Is a triangle with sides 7, 24 and 25 right-angled? Show your working.

**$7^2 + 24^2 = 49 + 576 = 625 = 25^2$. Yes, it is right-angled.**

**5** Paper investigation: measure a sheet of A4. Divide the long side by the short side. Fold it in half and do the same for the new A5 size.

A4: **297 ÷ 210 ≈ 1.41** **A5:** **210 ÷ 148 ≈ 1.42**

**Both are about √2 = 1.414…, so the shape stays the same.**


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GEOMETRY QUEST · WORLD 16

# Fractals

**ANSWER SHEET · for the grown-up**

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

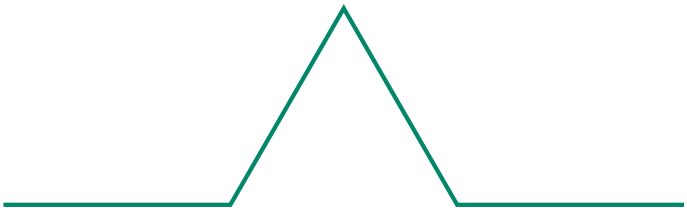
**1** Cantor set: each step removes the middle third of every piece. Complete the table.

| step | pieces | length left (start = 1) |
|---|---|---|
| 0 | **1** | **1** |
| 1 | **2** | **2/3** |
| 2 | **4** | **4/9** |
| 3 | **8** | **8/27** |
| 4 | **16** | **16/81** |

**2** Koch snowflake: start with a triangle of side 9 cm. Each step, every edge becomes 4 edges, each a third as long. Complete the table.

| step | edges | perimeter |
|---|---|---|
| 0 | **3** | **27 cm** |
| 1 | **12** | **36 cm** |
| 2 | **48** | **48 cm** |
| 3 | **192** | **64 cm** |

**3** Draw step 1 of the Koch curve on this 9 cm line: split it into three 3 cm parts, draw an equilateral triangle on the middle part, then rub out its base.



**4** Fractal hunt: find three things in nature or at home that repeat their pattern when you zoom in. Draw one of them.

**e.g. fern leaf, broccoli/cauliflower, tree branches, river deltas, frost, lightning.**


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